

Basics of complex variables

Consider the complex variable y

$$y = a + ib$$

where both a & b are real

$$\operatorname{Re}(y) = a \quad \leftarrow \text{real part}$$

$$\operatorname{Im}(y) = b \quad \leftarrow \text{imaginary part}$$

The "complex conjugate" of y is

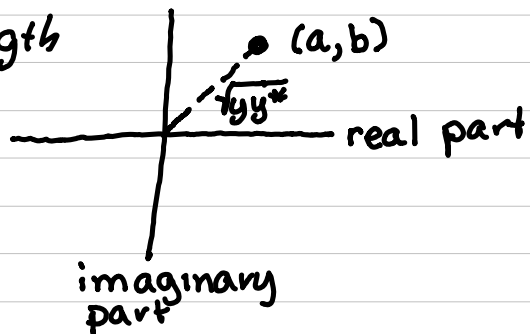
$$y^* = a - ib$$

so:

$$yy^* = [\operatorname{Re}(y)]^2 + [\operatorname{Im}(y)]^2 = a^2 + b^2$$

(remember, $i^2 = -1$)

the squared length
of a vector
in the
"complex
plane"



Exponentials

$$e^{ix} = \cos x + i \sin x$$

$$\operatorname{Re}[e^{ix}] = \cos x$$

$$\operatorname{Im}[e^{ix}] = \sin x$$

$$[e^{ix}][e^{ix}]^* = \cos^2 x + \sin^2 x = 1$$

ck this:

$$e^{ix} e^{-ix} = 1 \checkmark$$

Dirac delta function $\delta(x)$

A continuous distribution such that

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

and $\int_{-\infty}^{\infty} \delta(x) dx = 1$

and $\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$

and $\delta(x-a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iq(x-a)} dq$

More useful for us

Kronecker delta function:

A discrete version of the Dirac delta

$$\delta_{j,k} = \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases}$$

$$\sum_k \delta_{jk} a_k = a_j$$

$$\delta_{nm} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N e^{2\pi i k (n-m)/N}$$

Now let's consider a special case that will be useful to us.

say: $N = L/\Delta x$, so $N \rightarrow \infty$ as $\Delta x \rightarrow 0$

also, say $q = \frac{n\omega\pi}{L} = \frac{2\pi\Delta x n}{N}$ and $q' = \frac{2\pi\Delta x m}{N}$

$$\delta_{qq'} = \lim_{\Delta x \rightarrow 0} \frac{1}{L} \sum_{k=1}^N \Delta x e^{ik\Delta x (q-q')} \quad \text{now let's say } x_k = k\Delta x$$

$$= \frac{1}{L} \lim_{\Delta x \rightarrow 0} \sum_{x_k = \Delta x}^L e^{ix_k (q-q')} \quad \text{which can be converted to an integral}$$

$$\delta_{qq'} = \frac{1}{L} \int_0^L dx e^{ix(q-q')} \quad \text{we use this to get the Fourier spectrum!}$$